

Cycle Detection & Bipartite

Graphs, Cycle Detection, Bipartite Check · Mentor-Led Lab (Student Edition)

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Difficulty ★★★ Advanced

Problem

Analyse an undirected graph two ways: does it contain a cycle, and is it bipartite (two-colourable with no edge joining same colours)?

Implement undirected cycle detection (DFS with a parent) and the 2-colouring check (BFS) from scratch.

What to Compute

- Has Cycle: Yes if the graph contains a cycle, otherwise No.
- Bipartite: Yes if the graph can be 2-coloured with no edge between same colours, otherwise No.

Input / Output Format

Input: Line 1: N M. Next M lines: an undirected edge u v.

Output: Two lines: Has Cycle and Bipartite.

Examples

Example 1

Input: 4 3 0 1 1 2 2 0

Output:

Has Cycle: Yes
Bipartite: No

Explanation: The triangle 0-1-2 is an odd cycle, so the graph has a cycle and cannot be two-coloured.

Example 2

Input: 3 2 0 1 1 2

Output:

Has Cycle: No
Bipartite: Yes

Explanation: A path has no cycle and is bipartite (colour the ends one colour and the middle the other).

Constraints

- Cycle detection is for an undirected graph (ignore the edge back to the parent).
- Bipartite means a valid 2-colouring exists.
- Process disconnected graphs by restarting from each unvisited vertex.

Sample Run

Sample Input

```
4 3
0 1
1 2
2 0
```

Sample Output

```
Has Cycle: Yes
Bipartite: No
```